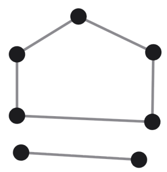
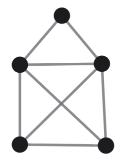
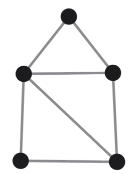
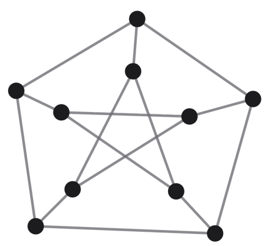
Math 55 worksheet [Monday/Wednesday] [MM]/[DD]/2026

1. Determine which of the following graphs have spanning trees.

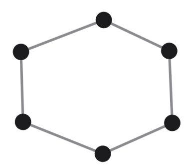








Petersen Graph



2. For the graphs above that do have spanning trees, find a spanning tree for each of them and 2-color it.

3. Let $\mathcal{C}$ be the set of all finite binary sequences, including the empty sequence $()$. Let $S \subseteq \mathcal{C}$. Consider the graph $G = (V, E)$ where $V = S$ and for any two sequences s, t, we have $\{s, t\} \in E$ iff either s extends t by one bit or t extends s by one bit. So for example, there is an edge between 1001 and 10010 but not between 1001 and 01010 (the empty string $()$ is just adjacent to 0 or 1, if either of them are in S).

   Prove that if S contains $()$ and is closed downward (i.e. if $t \in S$ and s is the first n bits of t for some n less than the length of t, then $s \in S$), then G is a tree.

4. Using the definitions from problem 3, prove that if S contains $()$, then G is a tree iff S is closed downward. (you just proved one of the directions)